Statistical Modelling of Stock Returns: A Summary
ADANI POWERS (ADANIPOWERS.NS) - 2023

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Which distribution best models the returns?

Based on a Kolmogorov-Smirnov (KS) goodness-of-fit test, the **Student's t-distribution** provides the most plausible model for the daily log returns of ADANI POWERS (ADANIPOWERS.NS) for the year 2023.

This conclusion is drawn because the Student's t-distribution yielded the highest p-value among all tested models, indicating the strongest statistical evidence of a good fit. The superiority of the t-distribution over the Normal distribution suggests that the stock's returns exhibit *leptokurtosis*, or "heavy tails." This means that extreme price movements (both positive and negative) are more common than a Normal distribution would predict, a well-documented characteristic of financial market data.

What are the estimated parameters?

The parameters for the best-fitting model, the Student's t-distribution, were estimated using Maximum Likelihood Estimation (MLE). The estimated parameters are:

- **Degrees of Freedom (ν):** 8.7941
- **Location / Mean (μ):** 0.0015
- **Scale / Std. Deviation (σ):** 0.0289

The low value for the degrees of freedom ($\nu < 30$) further confirms the presence of heavy tails in the return distribution.

What are the test results and p-values?

The goodness-of-fit for five candidate distributions was assessed using the Kolmogorov-Smirnov test. The null hypothesis (H_0) for each test was that the daily returns follow the specified distribution. The table below ranks the distributions based on their p-values.

Table 1: Kolmogorov-Smirnov Goodness-of-Fit Test Results

Distribution	**KS Statistic**	**p-value**
Student's t	0.0613372	0.30509
Laplace	0.0721945	0.149635
Logistic	0.0615026	0.302075
Normal	0.0716681	0.155324
Cauchy	0.0872343	0.045884

A higher p-value suggests that there is not enough evidence to reject the null hypothesis. With a p-value of **0.30509**, the **Student's t-distribution** stands out as the most appropriate model.

What do the confidence interval outliers indicate?

Assuming the returns follow a Student's t-distribution, the 95% confidence interval for daily log returns was calculated to be [**-0.064213, 0.067119**]. Returns that fall outside this range are considered outliers, representing days of statistically unusual price movements.

These outliers indicate that a significant market event likely occurred, causing volatility to exceed its typical range. Such events could include major corporate announcements, unexpected macroeconomic news, or significant shifts in market sentiment.